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Experiment No: 01

Experiment Name: Preparation and Testing of Straight-Through UTP Cable Using RJ45 Connectors

Theory: A Straight-Through UTP (Unshielded Twisted Pair) cable is an Ethernet cable used to connect different network devices, such as a computer to a switch or a router. In this cable, both ends are wired using the same standard (usually T568B). The cable consists of eight color-coded wires arranged inside an RJ45 connector. After crimping the connectors, a LAN cable tester is used to check whether all eight wires are connected correctly, and the cable is functioning properly.

Instruments Required

1. UTP (Cat5e/Cat6) Cable
2. RJ45 Connectors (2 pieces)
3. RJ45 Crimping Tool
4. Wire Cutter/Stripper
5. LAN Cable Tester
6. Scissors or Cable Stripper (optional)

Procedure

1. Cut the UTP cable to the required length using a wire cutter.
2. Remove approximately 2–3 cm of the outer jacket from both ends of the cable without damaging the inner wires.
3. Untwist the four wire pairs and arrange the eight wires according to the T568B color sequence (White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown).

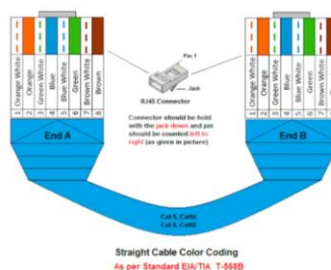


Figure 1: Wire Arrangement

4. Straighten the wires and trim them so that all conductors are of equal length.
5. Carefully insert the arranged wires into the RJ45 connector, ensuring that each wire reaches the end of the connector and remains in the correct order.

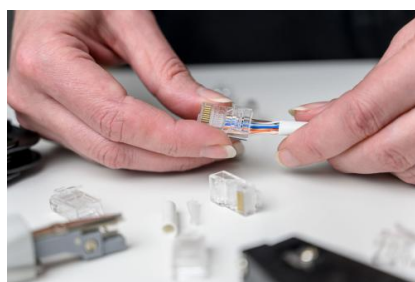


Figure 2: Inserting Wires into the RJ45 Connector

6. Place the RJ45 connector into the crimping tool and press the handles firmly to secure the connector and establish proper electrical contact.



Figure 3: Crimping the RJ45 Connector

7. Repeat the same process on the other end of the cable using the same T568B wiring sequence to create a straight-through cable.
8. Connect both ends of the completed cable to a LAN cable tester and switch it on.
9. Observe the LED indicators. If LEDs 1 through 8 light up sequentially on both the master and remote units, the cable has been prepared successfully.



Figure 4: LAN Cable Tester Showing Correct Result

Result: A straight-through UTP cable was successfully prepared using the T568B wiring standard. The LAN cable tester confirmed proper continuity by displaying LEDs 1–8 in the correct sequence on both the master and remote units, indicating that the cable is functioning properly.

Precautions

1. Use the same wiring standard (T568B) on both ends of the cable.
2. Avoid excessive untwisting of the wire pairs.
3. Ensure all wires are fully inserted before crimping.
4. Trim the wires evenly for proper insertion.
5. Apply sufficient pressure while crimping the connector.
6. Check the cable with a LAN cable tester before use.
7. Handle the RJ45 connectors carefully to prevent damage.

This format is the standard style used in engineering laboratory reports, where figures are referenced naturally within the procedure instead of being listed as separate sections.

Experiment No: 02**Experiment Name:** Preparation and Testing of Crossover UTP Cable Using RJ45 Connectors**Theory**

A Crossover UTP (Unshielded Twisted Pair) cable is used to connect similar network devices directly, such as computer-to-computer, switch-to-switch, or hub-to-hub, without using an intermediate networking device. One end of the cable is wired using the T568A standard, while the other end is wired using the T568B standard. After crimping the RJ45 connectors, a LAN cable tester is used to verify the correctness of the cable connections.

Instruments Required

- UTP (Cat5e/Cat6) Cable
- RJ45 Connectors (2 pieces)
- RJ45 Crimping Tool
- Wire Cutter/Stripper
- LAN Cable Tester

Procedure

1. Cut the UTP cable to the required length using a wire cutter.
2. Strip approximately 2–3 cm of the outer jacket from both ends of the cable.
3. Untwist the wire pairs and arrange the wires at one end according to the T568A standard and the other end according to the T568B standard.

End A (T568A):

1. White/Green
2. Green
3. White/Orange
4. Blue
5. White/Blue
6. Orange
7. White/Brown
8. Brown

End B (T568B):

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

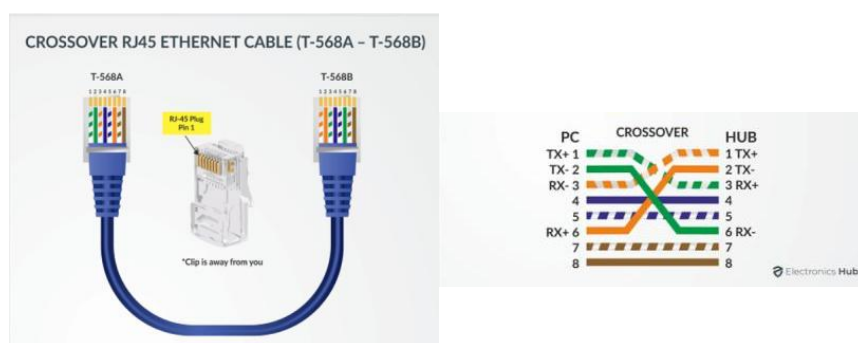


Figure1: Wire Arrangement (T568A and T568B)

4. Straighten the wires and trim them evenly.
5. Insert the arranged wires into the RJ45 connector carefully, ensuring that each wire reaches the front of the connector and remains in the correct order.

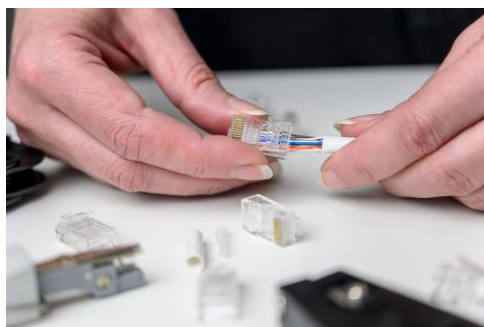


Figure2: Wire Insertion into RJ45 Connector

6. Place the RJ45 connector into the crimping tool and press firmly to crimp the connector onto the cable.



7. Repeat the same process on the other end using the appropriate wiring standard (T568A or T568B) to complete the crossover cable.
8. Connect both ends of the cable to a LAN cable tester and switch it on.
9. Observe the LED sequence. If the tester indicates the correct crossover wiring (where the transmit and receive pairs are crossed), the cable has been prepared successfully.



Result: A Crossover UTP cable was successfully prepared by terminating one end with the T568A wiring standard and the other end with the T568B wiring standard. The LAN cable tester confirmed that the cable connections were correct, indicating that the crossover cable is functioning properly.

Precautions

1. Ensure one end is wired using T568A and the other end using T568B.
2. Do not untwist the wire pairs more than necessary.
3. Trim all wires to the same length before inserting them into the RJ45 connector.
4. Make sure each wire reaches the end of the RJ45 connector before crimping.
5. Apply firm and even pressure while crimping.
6. Verify the cable using a LAN cable tester after crimping.
7. Handle the RJ45 connectors carefully to avoid damaging the metal contacts.

Experiment No: 03

Experiment Name: Preparation and Testing of Crossover UTP Cable Using RJ45 Connectors

Theory: A Bus Topology is a network topology in which all devices are connected through a single communication line called the backbone. Data transmitted by one device travels along the backbone and is received by all connected devices, but only the intended destination processes the data. Bus topology is simple, cost-effective, and suitable for small networks. Cisco Packet Tracer allows users to design, configure, and simulate a bus topology without using physical networking equipment.

Instruments / Software Required

- Computer or Laptop
- Cisco Packet Tracer
- Hub
- PCs (End Devices)
- Copper Straight-Through Cable

Procedure

1. Open Cisco Packet Tracer and create a new project.
2. Place one Hub and four PCs on the workspace.
3. Connect each PC to the Hub using Copper Straight-Through cables.
4. Configure the IP addresses of all PCs so that they belong to the same network.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

5. Verify that all cable connections are active and the network topology is complete.

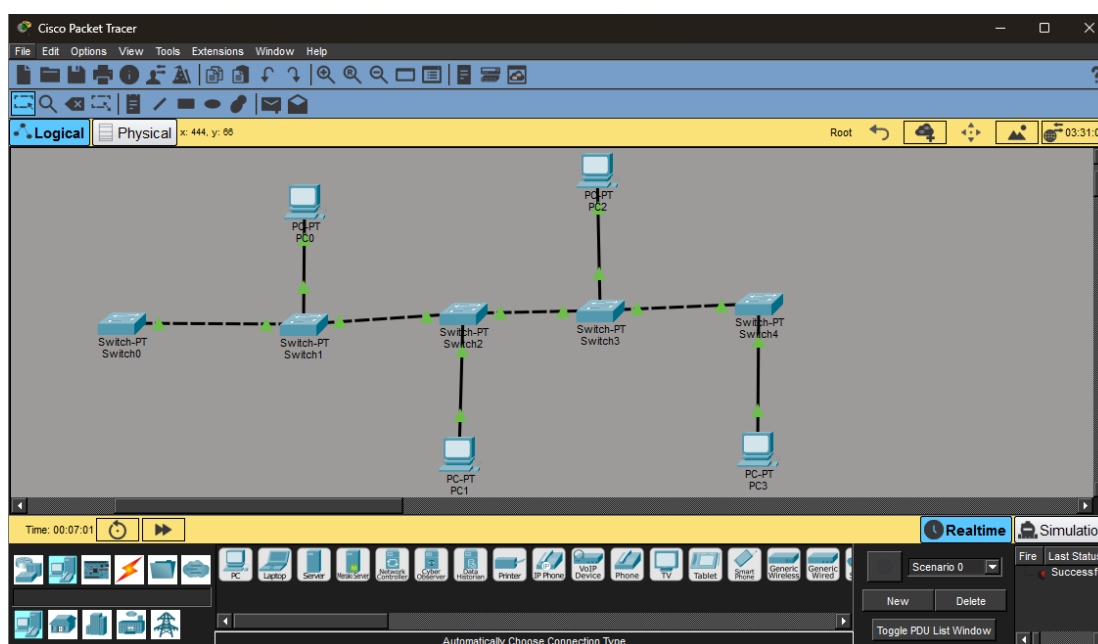


Figure1: Bus Topology in Cisco Packet Tracer

6. Open the Command Prompt on one PC and use the ping command to test communication with the other PCs.
7. Confirm that all ping requests receive successful replies, indicating proper network connectivity.

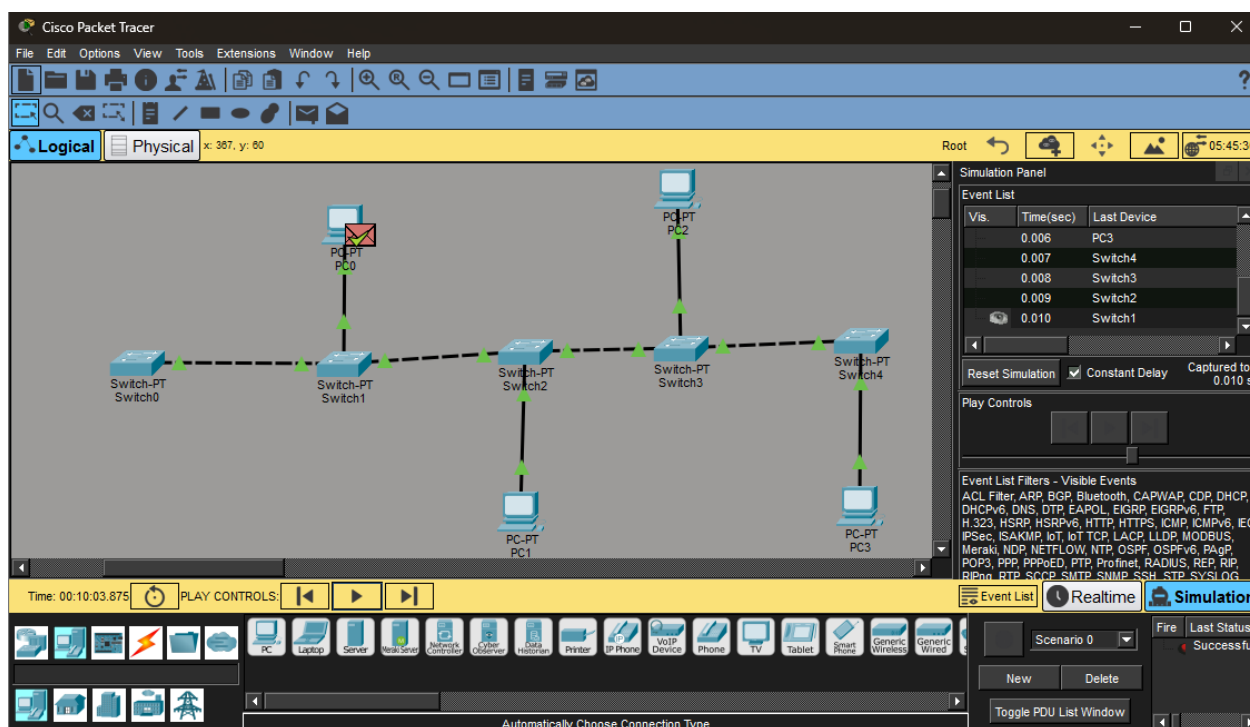


Figure 2: Successful Simulation Output

Result

The Bus Topology was successfully designed and implemented using Cisco Packet Tracer. All computers were connected through the hub, assigned valid IP addresses, and communicated successfully. The successful ping results confirmed that the network was configured correctly.

Precautions

1. Use the correct cable type when connecting devices.
2. Assign unique IP addresses to each PC.
3. Ensure all devices are in the same subnet.
4. Verify that all cable connections are active before testing.
5. Avoid duplicate IP addresses in the network.
6. Save the Packet Tracer project after completing the simulation.
7. Check the ping results to ensure successful communication between all devices.

Experiment No: 04**Experiment Name:** Design and Implementation of Ring Topology Using Cisco Packet Tracer**Theory**

A Ring Topology is a network topology in which each device is connected to two neighboring devices, forming a closed loop or ring. Data travels from one device to another around the ring until it reaches the destination. Ring topology provides an organized method of data transmission and is suitable for networks where orderly communication is required. Cisco Packet Tracer is used to design and simulate the ring topology without the need for physical networking devices.

Instruments / Software Required

- Computer or Laptop
- Cisco Packet Tracer
- Switches (or devices as instructed)
- PCs (End Devices)
- Copper Straight-Through Cable

Procedure

1. Open Cisco Packet Tracer and create a new project.
2. Place the required Switches and PCs on the workspace.
3. Connect the switches in a ring structure, ensuring the last switch is connected back to the first switch.
4. Connect each PC to its respective switch using Copper Straight-Through cables.
5. Configure the IP addresses of all PCs so that they belong to the same network.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

6. Verify that all cable connections are active and the ring topology has been completed.

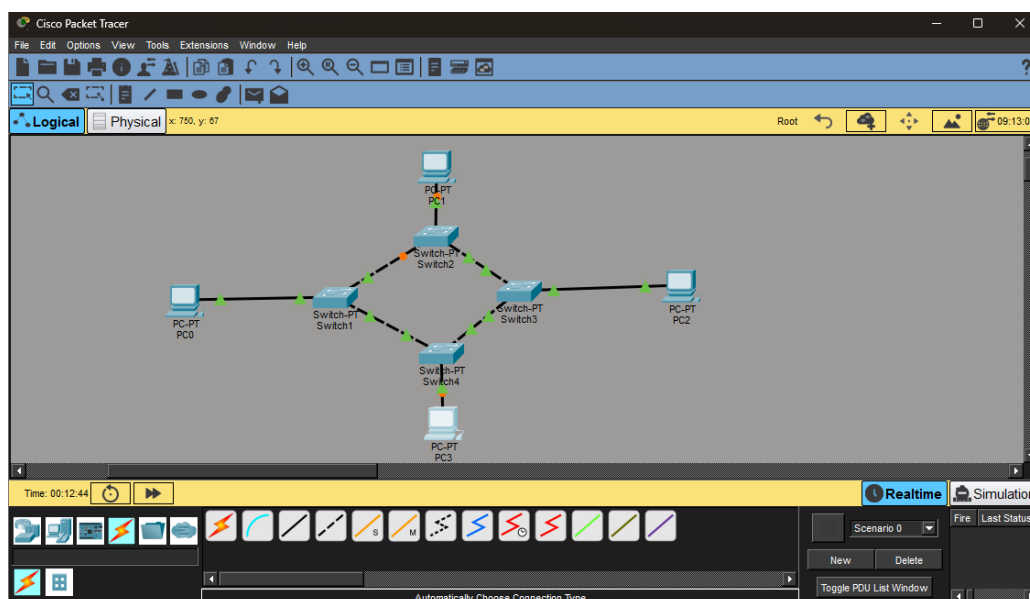


Figure 1: Ring Topology in Cisco Packet Tracer

7. Open the Command Prompt on one PC and use the ping command to test communication with the other PCs.
8. Verify that all ping requests receive successful replies, confirming that the ring network is functioning correctly.

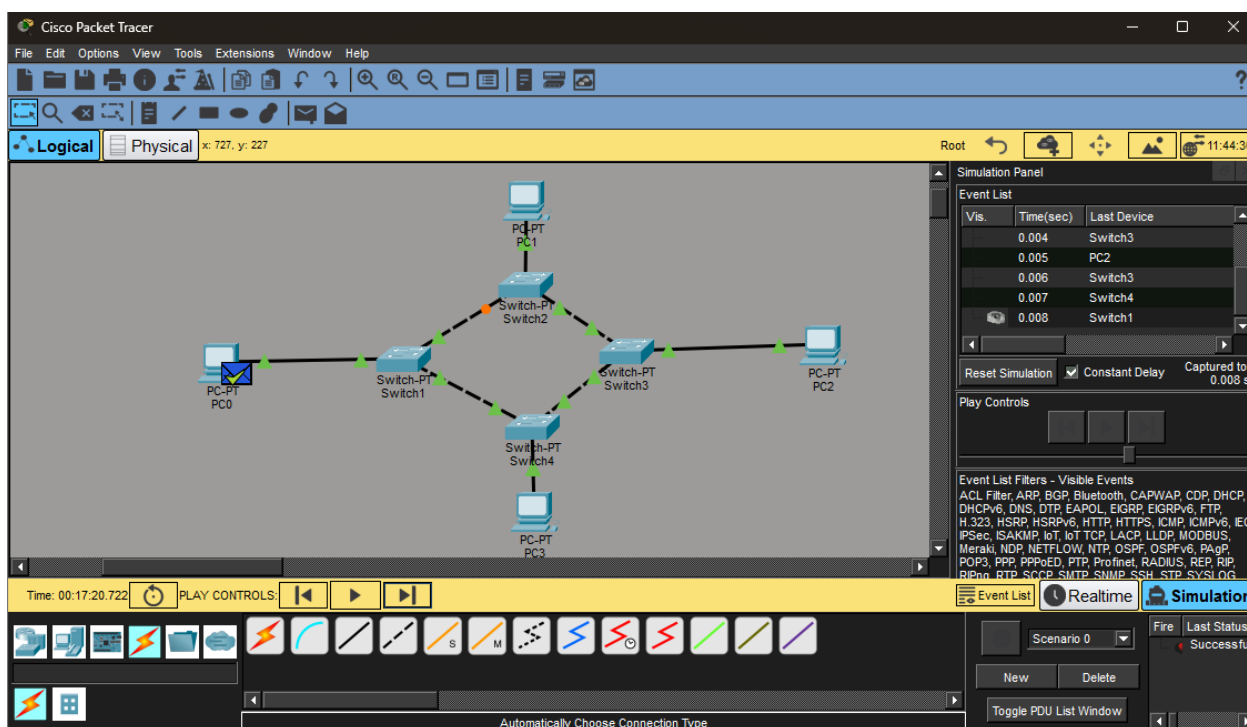


Figure 2: Successful Simulation Output

Result

The Ring Topology was successfully designed and implemented using Cisco Packet Tracer. All devices were connected in a ring configuration, assigned valid IP addresses, and communicated successfully. The successful ping results confirmed that the network was configured correctly.

Precautions

1. Ensure the ring connection is complete by connecting the last switch back to the first switch.
2. Use the correct cable type for all connections.
3. Assign unique IP addresses to each PC.
4. Verify that all devices are in the same subnet.
5. Check that all cable links are active before testing.
6. Save the Packet Tracer project after completing the simulation.
7. Test communication using the ping command to confirm proper network connectivity.

Experiment No: 05

Experiment Name: Design and Implementation of Star Topology Using Cisco Packet Tracer

Theory: A Star Topology is a network topology in which all devices are connected to a central device, such as a switch or hub. All communication between devices passes through the central device. It is one of the most commonly used network topologies because it is easy to install, manage, and troubleshoot. If one cable or device fails, the rest of the network continues to operate normally. Cisco Packet Tracer is used to design and simulate a star topology without physical networking equipment.

Instruments / Software Required

- Computer or Laptop
- Cisco Packet Tracer
- Switch
- PCs (End Devices)
- Copper Straight-Through Cable

Procedure

1. Open Cisco Packet Tracer and create a new project.
2. Place one Switch and four PCs on the workspace.
3. Connect each PC to the switch using Copper Straight-Through cables.
4. Configure the IP addresses of all PCs so that they belong to the same network.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

5. Verify that all cable connections are active and the star topology is complete.

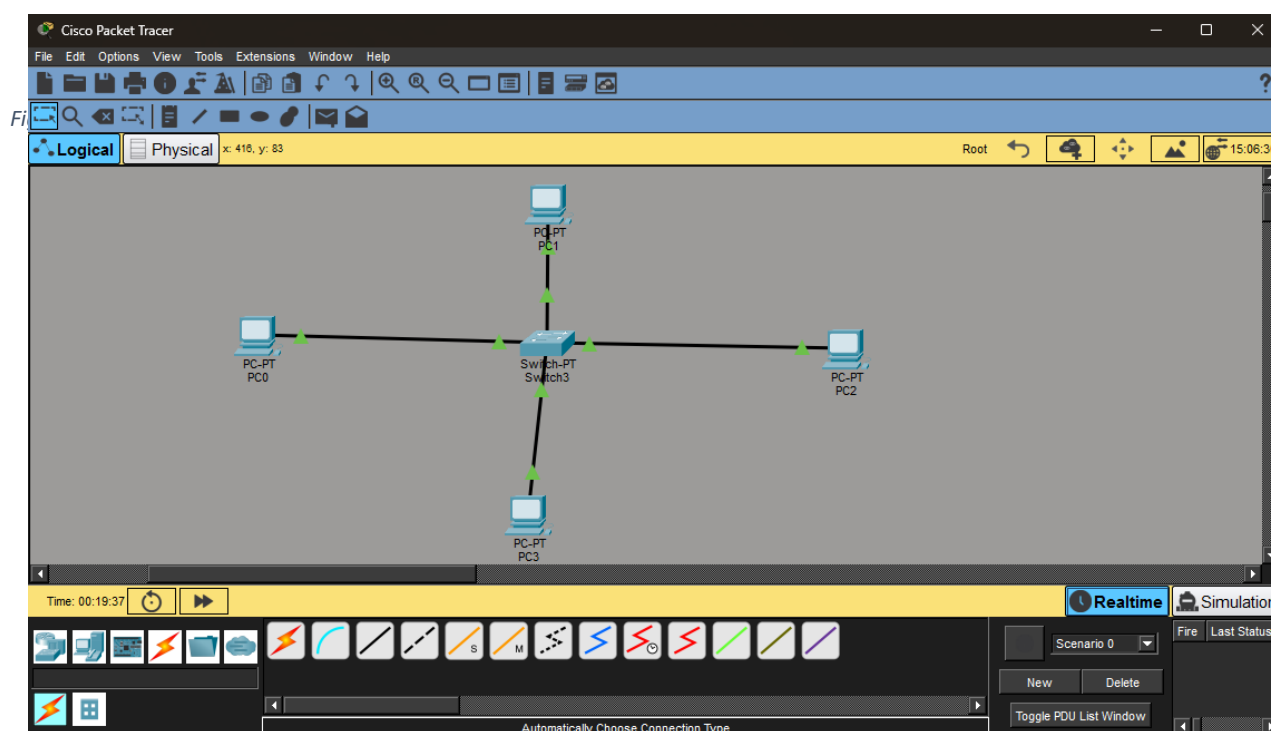


Figure 1: Star Topology in Cisco Packet Tracer

6. Open the Command Prompt on one PC and use the ping command to test communication with the other PCs.
7. Ensure that all ping requests receive successful replies, confirming proper network connectivity.

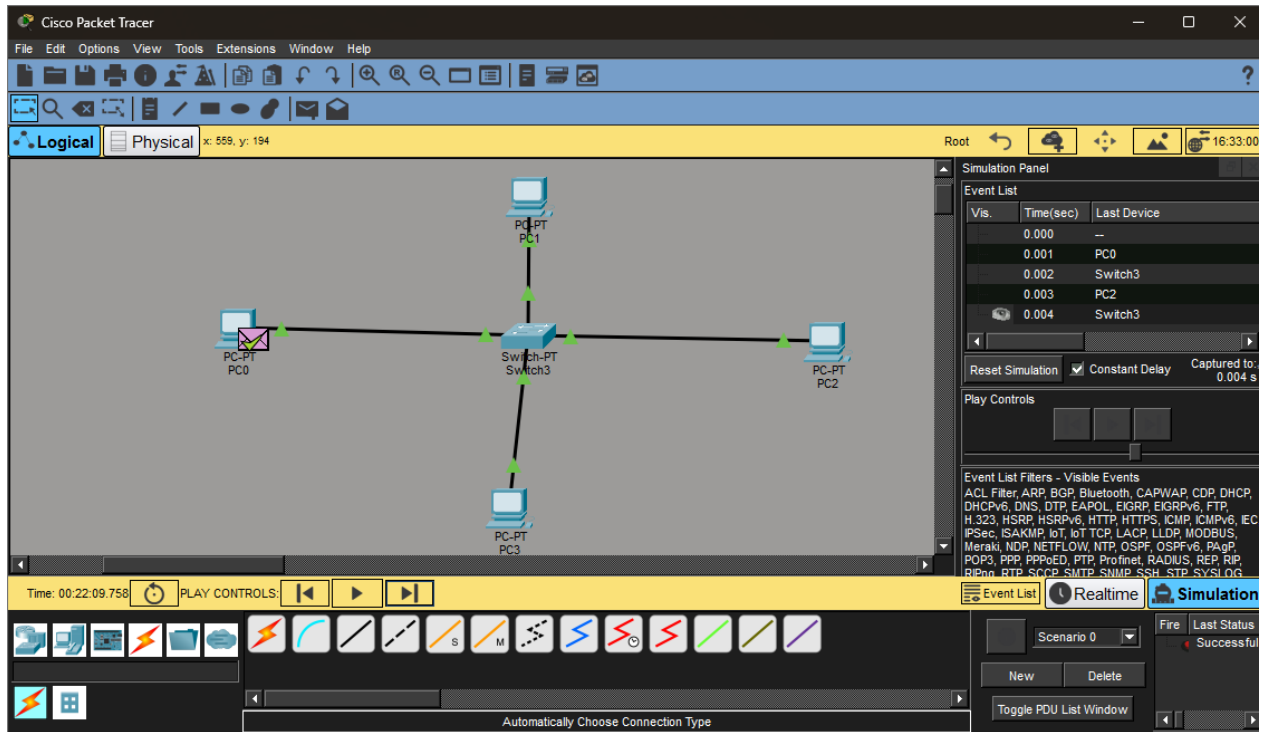


Figure 2: Successful Simulation Output

Result

The Star Topology was successfully designed and implemented using Cisco Packet Tracer. All PCs were connected to a central switch, assigned valid IP addresses, and communicated successfully. The successful ping results confirmed that the network was configured and operating correctly.

Precautions

1. Use Copper Straight-Through cables to connect each PC to the switch.
2. Assign a unique IP address to each PC.
3. Ensure all devices are configured within the same subnet.
4. Verify that all cable connections are active before testing.
5. Avoid duplicate IP addresses in the network.
6. Save the Packet Tracer project after completing the simulation.
7. Test the network using the ping command to verify successful communication.

Experiment No: 06

Experiment Name: Design and Implementation of Tree Topology Using Cisco Packet Tracer

Theory: A Tree Topology is a hierarchical network topology that combines the characteristics of Star and Bus topologies. It consists of a root or central device connected to multiple secondary devices, forming a tree-like structure. This topology is suitable for large networks because it is scalable, easy to expand, and simplifies network management. Cisco Packet Tracer allows users to design and simulate a tree topology without physical networking equipment.

Instruments / Software Required

- Computer or Laptop
- Cisco Packet Tracer
- Switches
- PCs (End Devices)
- Copper Straight-Through Cable

Procedure

1. Open **Cisco Packet Tracer** and create a new project.
2. Place the required **Switches** and **PCs** on the workspace.
3. Connect the switches in a hierarchical (tree) structure, with one switch acting as the root switch and the others as branch switches.
4. Connect each PC to its respective switch using **Copper Straight-Through** cables.
5. Configure the IP addresses of all PCs so that they belong to the same network.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

6. Verify that all cable connections are active and the tree topology is complete.

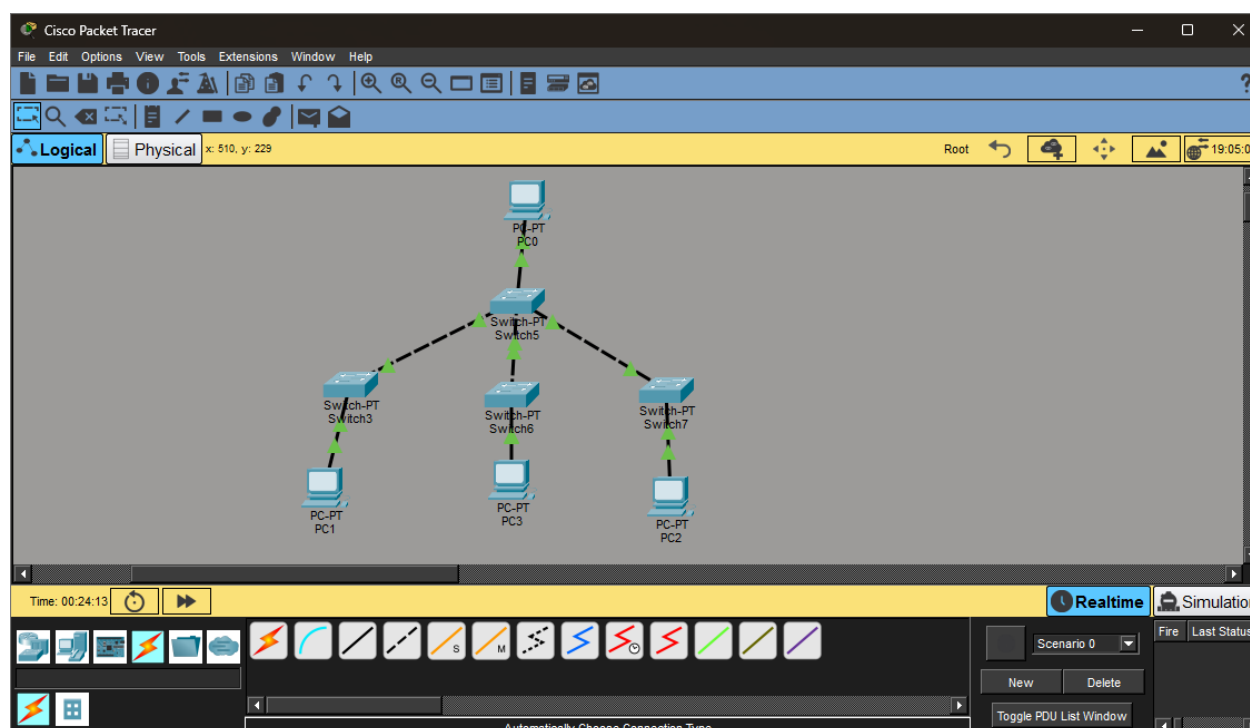


Figure 1: Tree Topology in Cisco Packet Tracer

7. Open the Command Prompt on one PC and use the ping command to test communication with the other PCs.
8. Verify that all ping requests receive successful replies, confirming that the network is functioning properly.

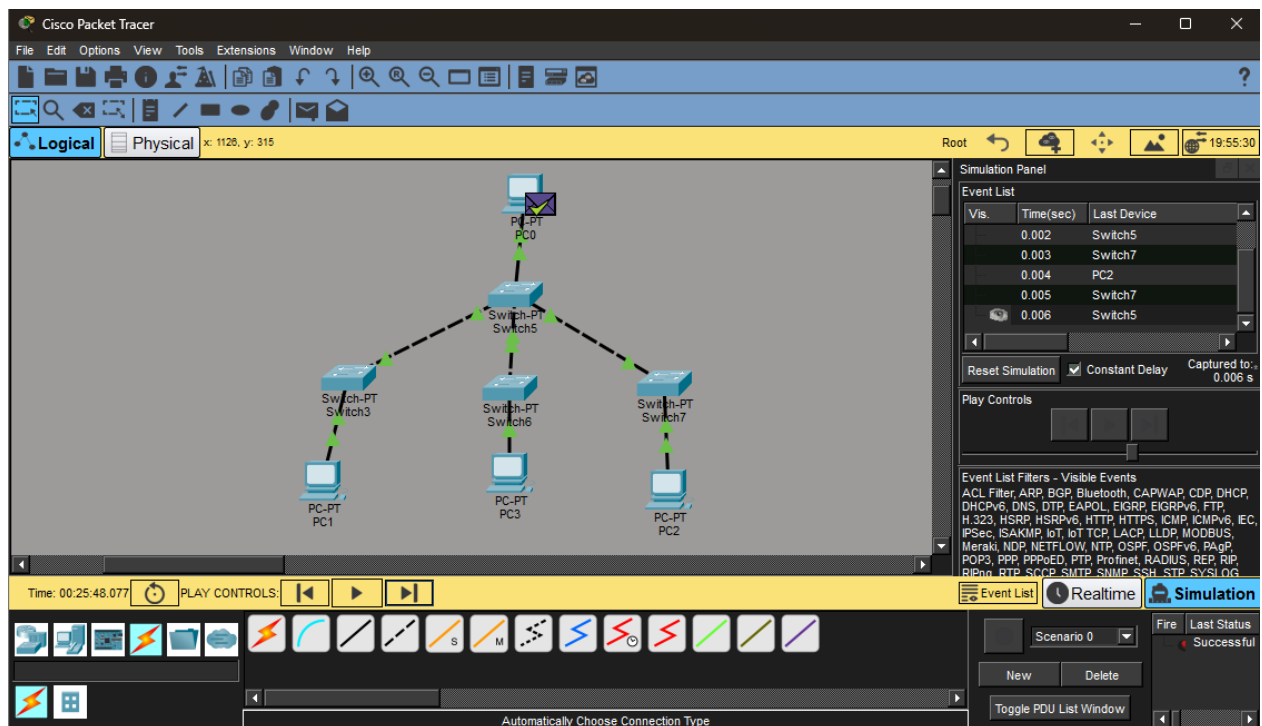


Figure 2: Successful Simulation Output

Result

The Tree Topology was successfully designed and implemented using Cisco Packet Tracer. All devices were connected in a hierarchical structure, assigned valid IP addresses, and communicated successfully. The successful ping results confirmed that the network was configured correctly and operated as expected.

Precautions

1. Connect the switches in the correct hierarchical structure.
2. Use Copper Straight-Through cables for all required connections.
3. Assign unique IP addresses to each PC.
4. Ensure all devices are configured within the same subnet.
5. Verify that all cable connections are active before testing.
6. Save the Packet Tracer project after completing the simulation.
7. Test network connectivity using the **ping** command to ensure proper communication.

Experiment No: 07**Experiment Name: Design and Implementation of Mesh Topology Using Cisco Packet Tracer**

Theory: A Mesh Topology is a network topology in which every device is directly connected to every other device in the network. This provides multiple communication paths, making the network highly reliable and fault-tolerant. Even if one connection fails, data can still be transmitted through another path. Cisco Packet Tracer is used to design and simulate a mesh topology without requiring physical networking equipment.

Instruments / Software Required

- Computer or Laptop
- Cisco Packet Tracer
- Switches/PCs (as required)
- Copper Straight-Through Cable
- Copper Cross-Over Cable (if required)

Procedure

1. Open Cisco Packet Tracer and create a new project.
2. Place the required Switches and PCs on the workspace.
3. Connect each device to every other device according to the mesh topology.
4. Connect the PCs to the appropriate switches using Copper Straight-Through cables.
5. Configure the IP addresses of all PCs so that they belong to the same network.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

6. Verify that all cable connections are active and the mesh topology is complete.

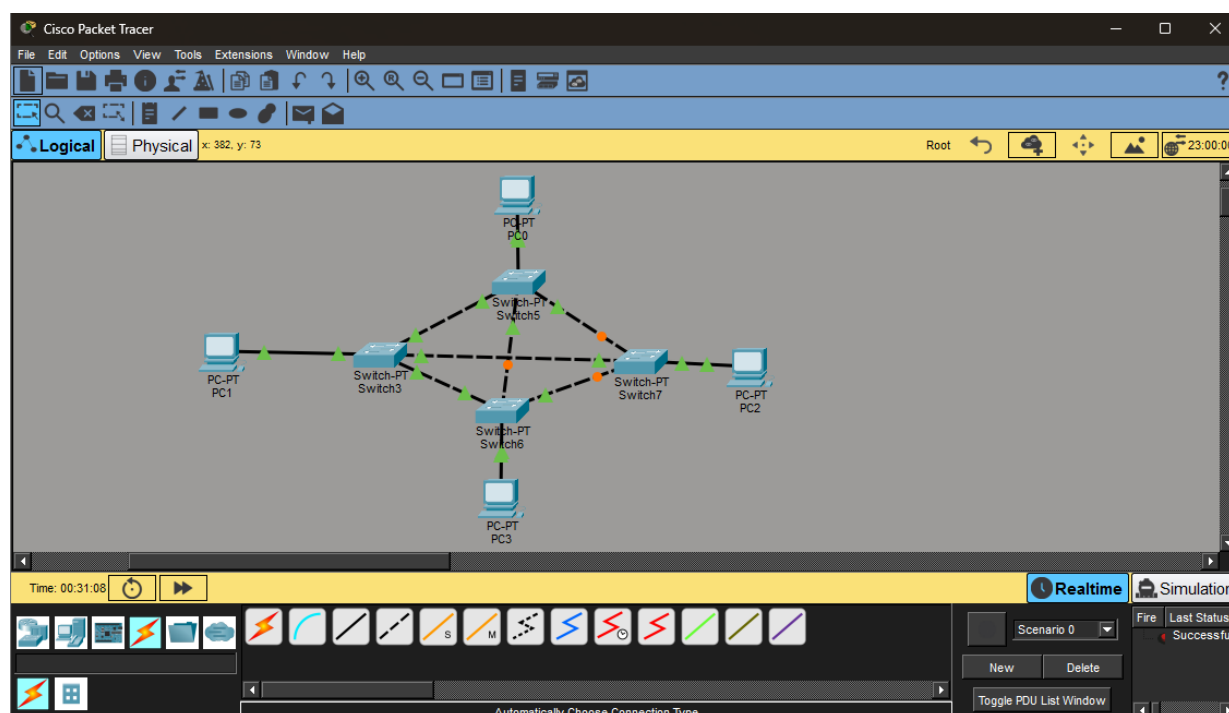


Figure 1: Mesh Topology in Cisco Packet Tracer

7. Open the Command Prompt on one PC and use the ping command to test communication with the other PCs.
8. Verify that all ping requests receive successful replies, confirming that the network is functioning properly.

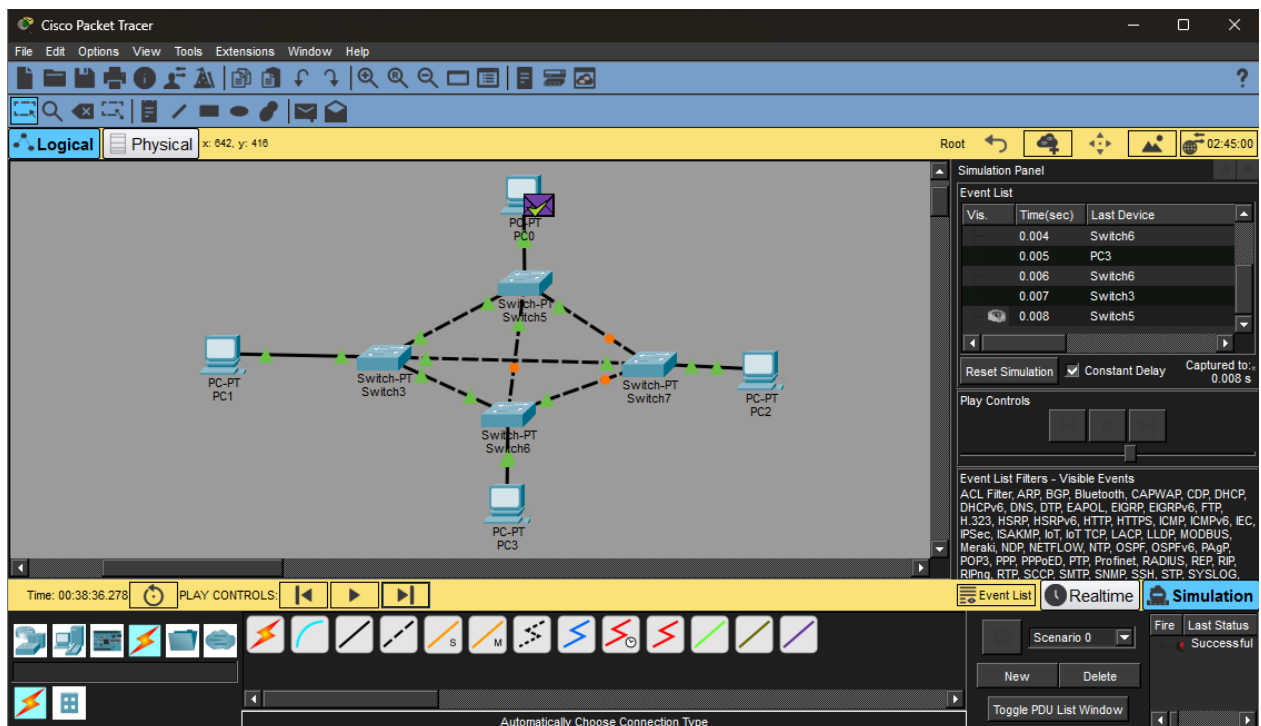


Figure 2: Successful Simulation Output

Result

The Mesh Topology was successfully designed and implemented using Cisco Packet Tracer. All devices were interconnected, assigned valid IP addresses, and communicated successfully. The successful ping results confirmed that the network was configured correctly and functioning properly.

Precautions

1. Ensure that all required device connections are established correctly.
2. Use the appropriate cable type for each connection.
3. Assign unique IP addresses to all PCs.
4. Configure all devices within the same subnet.
5. Verify that all cable links are active before testing.
6. Save the Packet Tracer project after completing the simulation.
7. Test the network using the ping command to verify successful communication.

Experiment No: 08**Experiment Name:** Design and Implementation of Hybrid Topology Using Cisco Packet Tracer

Theory: A Hybrid Topology is a network topology that combines two or more different topologies, such as Star, Bus, Ring, or Mesh, into a single network. It offers greater flexibility, scalability, and reliability, making it suitable for large organizations and enterprise networks. Cisco Packet Tracer allows users to design, configure, and simulate hybrid networks without using physical networking devices.

Instruments / Software Required

- Computer or Laptop
- Cisco Packet Tracer
- Switches and/or Hubs
- PCs (End Devices)
- Copper Straight-Through Cable
- Copper Cross-Over Cable (if required)

Procedure

1. Open Cisco Packet Tracer and create a new project.
2. Place the required Switches, Hubs, and PCs on the workspace.
3. Design a Hybrid Topology by combining two or more network topologies (e.g., Star-Bus or Star-Ring).
4. Connect the network devices using the appropriate cables.
5. Configure the IP addresses of all PCs so that they belong to the same network.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

6. Verify that all cable connections are active and the hybrid topology has been completed successfully.

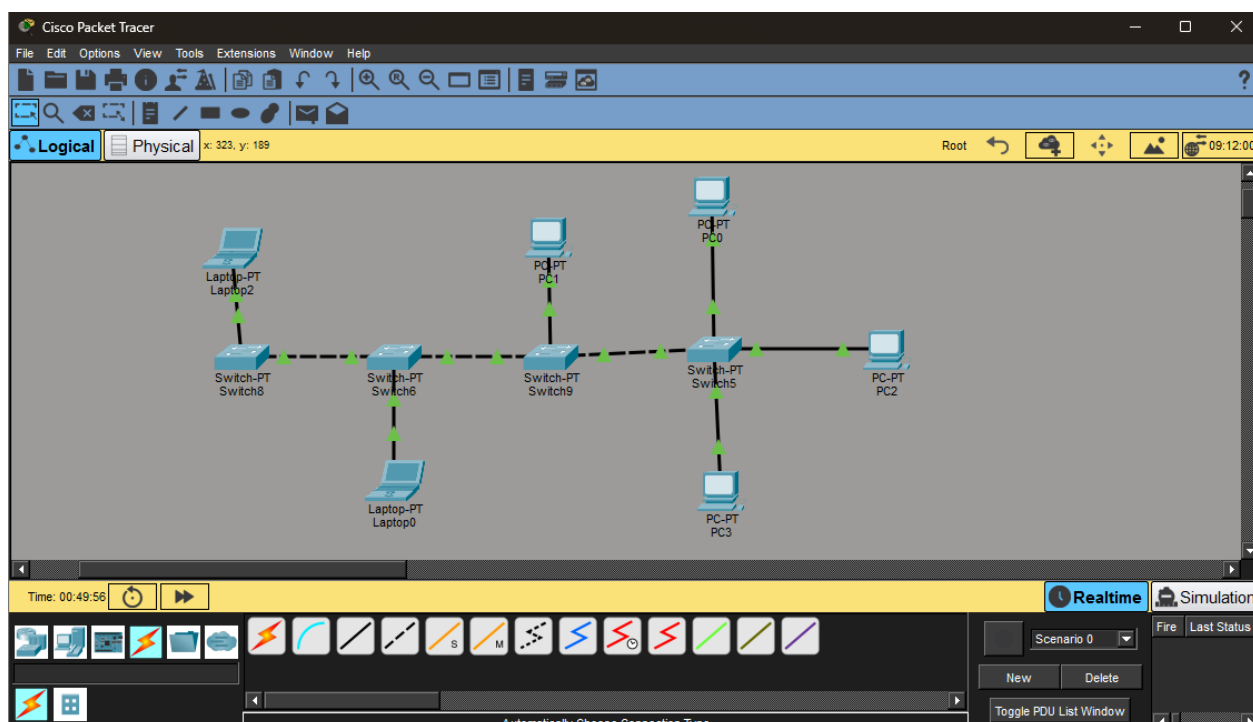


Figure 1: Hybrid Topology in Cisco Packet Tracer

7. Open the Command Prompt on one PC and use the ping command to test communication with the other PCs.
8. Confirm that all ping requests receive successful replies, indicating that the hybrid network is functioning correctly.

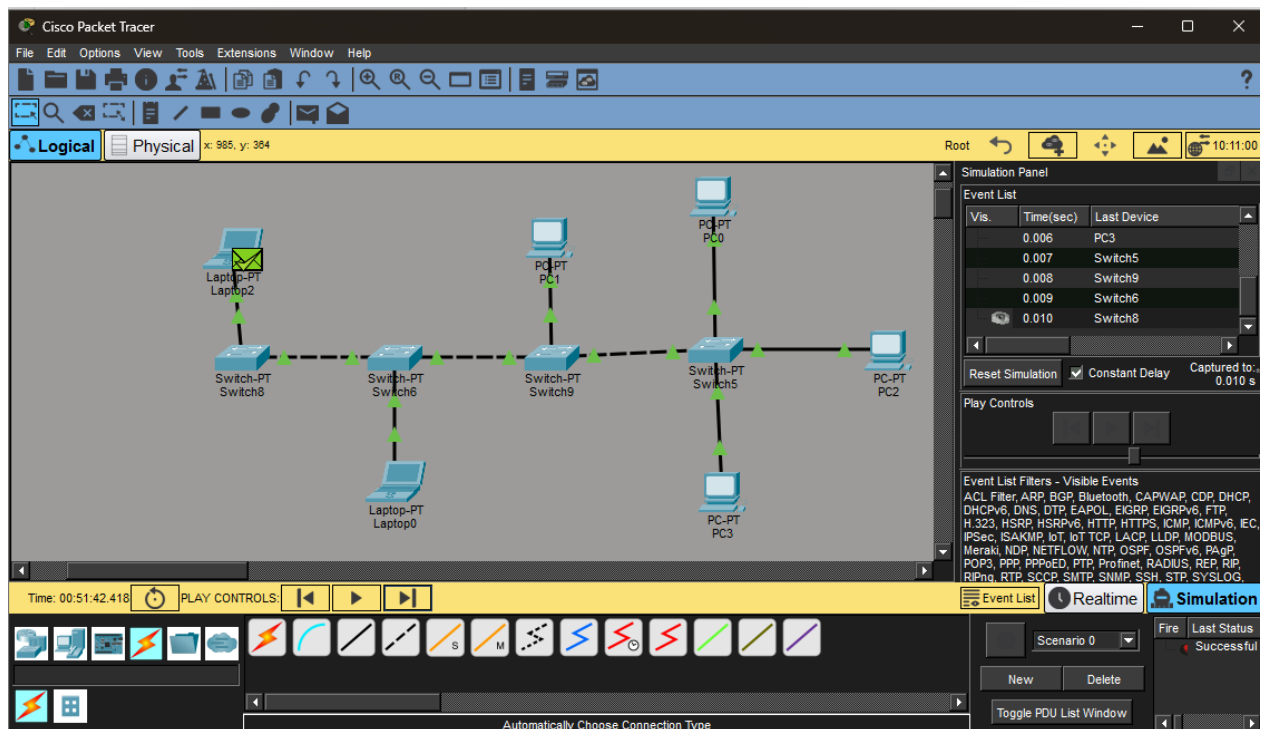


Figure 2: Successful Simulation Output

Result

The Hybrid Topology was successfully designed and implemented using Cisco Packet Tracer. Different network topologies were combined into a single network, and all devices were assigned valid IP addresses. The successful ping results confirmed that the network was configured correctly and communication between devices was successful.

Precautions

1. Design the hybrid topology according to the required network structure.
2. Use the appropriate cable type for each device connection.
3. Assign unique IP addresses to all PCs.
4. Ensure that all devices are configured within the same subnet.
5. Verify that all cable connections are active before testing.
6. Save the Packet Tracer project after completing the simulation.
7. Test the network using the ping command to ensure successful communication between all devices.